



Darstellung des Themas und Skizzierung des Forschungsvorhabens

Description of the topic and outline of the research project

Anlage zum Antrag auf Zulassung zum Promotionsverfahren in der MIN-Fakultät

Attachment to the application for admission to doctoral procedures at the MIN Faculty

A. Antrag | *Application*

Name, Vorname der Antragstellerin bzw. des Antragstellers | *Surname, first name of applicant:*

Damerow, Hauke

Vorläufiger Arbeitstitel (in Deutsch oder Englisch) | *Working title (in German or English):*

Novel Ab-Initio Methodology for Trions and Biexcitons in Moiré and Janus 2D materials

Ort
Place

Datum
Date

Unterschrift der Antragstellerin bzw. des Antragstellers
Signature of applicant

B. Zustimmung der Betreuenden / der Betreuungskommission | *Supervisor's or supervision panel's consent*

Ich / wir befürworten die vorgelegte Skizzierung des Promotionsvorhabens.

I / we approve the outlined dissertation project.

Betreuerin bzw. Betreuer | *Supervisor*

Datum
Date

Name und Unterschrift der Betreuerin bzw. des Betreuers
Name and signature of supervisor

Co-Betreuerin bzw. Co-Betreuer | *Co-supervisor*

(falls zutreffend) | *(if applicable)*

Datum
Date

Name und Unterschrift der Co-Betreuerin bzw. des Co-Betreuers
Name and signature of co-supervisor

Vorsitzende bzw. Vorsitzender | *Chair*

(falls zutreffend) | *(if applicable)*

Datum
Date

Name und Unterschrift der Vorsitzenden bzw. des Vorsitzenden der Betreuungskommission bzw. einer weiteren Co-Betreuerin bzw. eines weiteren Co-Betreuers
Name and signature of chair of the supervision panel or a further co-supervisor



C. Zusammenfassung zum Thema und zum Forschungsvorhaben | *Outline of the topic and research project*

Die Zusammenfassung kann in Deutsch oder Englisch erstellt werden und soll maximal zwei Seiten (Arial 12pt) umfassen. Text bitte in das untere Feld eintragen oder an dieses PDF-Formular anhängen. Sofern Letzteres, dann das vorliegende Formular (vollständig unterschrieben) und die PDF-Datei mit der Zusammenfassung zu einer PDF-Datei zusammenfügen.

Please note that the outline can be written in German or English and should not exceed two pages (Arial, 12pt). Please enter outline in the field below or attach to this PDF form. If the latter, then merge the present form (fully signed) and the PDF file with the outline into one PDF file.

Low-dimensional 2D materials have attracted a huge interest from the international research community in recent years. They are considered a key class of materials for studying fundamental quantum mechanical effects, e.g. unconventional superconductivity. They are also a key material class for many potential future applications, ranging from more effective solar cells via faster computation, energy storage, up to quantum computing. 2D materials can be created by exfoliation from the bulk material; characteristic of them are the strong covalent bonds in the 2-dimensional layers and weaker van der Waals forces between the different layers.

The physical properties of 2D materials and their heterostructures differ strongly from those of their bulk material form.

Transition metal dichalcogenides (TMDs) form a subclass of 2D materials. They are considered very versatile, as they can have a direct band gap in the visible wavelength range and show strong spin-orbit coupling.

Since the formation of twisted 2D bilayer graphene, which is superconducting only at different 'magic' twist angles between the stacked and one slightly rotated graphene monolayers, there has been a growing research activity on these so-called Moiré materials. The twist angle is a new degree of freedom which allows significant changes in the electronic band structure. This discovery has founded the new, currently burgeoning field of twistrionics, offering the chance to develop new optoelectronic nanophotonic and spintronic applications.

In this PhD project, Moiré TMDs are studied theoretically.

In TMD bilayers, excitons, electron-hole pair quasiparticles, with high binding energies up to several 100 meV, can form due to the reduced screening and strong Coulomb interactions. This means that even at room temperature, stable long-lifetime excitons can form, and therefore the photonics of these materials is dominated by the excitons. There are various kinds of excitons which can form, be modelled and observed in TMD 2D bilayers: Dark and bright excitons, and interlayer and intralayer excitons.

This could be very versatile for new nanophotonic applications.



The formation of excitons in 2D materials has not been fully understood at the fundamental level so far. Also there are many open questions about the interaction of different excitons in 2D materials: with each other and with other quasiparticles, such as phonons, and about their interaction with light pulses.

The most precise modelling of quantum systems is done with first-principles ab initio methods.

The study of the exciton physics at different time scales, up to the attosecond range, with different ab initio methods is the focus of this PhD project. These methods will be mainly based on time-dependent Density Functional Theory (TD-DFT), the GW approximation and the Bethe-Salpeter formula.

My and our main interest in this PhD project are the formation, interaction, and light-matter interaction of neutral excitons, trions and biexcitons in Moiré and Janus 2D materials.

Two electron-hole pairs can bind together and form a so-called biexciton. Trions are charged complex excitons of one electron-hole pair and an additional electron or hole which is bound to this electron-hole pair, forming a negatively or positively charged quasiparticle. These quasiparticles can be modelled and studied with quantum many-body methods. As analytically accurate solutions are hardly possible, accurate approximations methods, such as DFT, are used.

In this PhD project, an existing Ab Initio computation code is adapted to expand it for the computation of Moiré 2D materials and to model simple and more complex excitons in 2D TMDs with it.

In the second half of the PhD project, simple and complex excitons and their interaction with ultrashort laser pulses are modelled and studied, such as HHG of trions in TMDs. The results are compared with experimental observations measured by groups at the Max Planck Institute of Structure and Dynamics of Matter or by external groups. The results should be published in high-ranked scientific journals.

The PhD project is integrated and will contribute new insights to the DFG priority program SPP 2244, "2D materials - Physics of van der Waals (hetero)structures. In this project, excited states in van der Waals heterostructures are studied with ab initio DFT methods and with semiempirical methods.

This PhD project will contribute to a better fundamental understanding of complex exciton formation and of the interaction inside the exciton and with other excitons and laser photons in this van der Waals heterostructure. This understanding can also help to develop advanced nanophotonic applications, such as excitonic LEDs or excitonic solar cells.